



The role of artificial intelligence in preventing corporate crime

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ABSTRACT

This article explores the novel and transformative role AI plays in enhancing corporate compliance and monitoring to mitigate corporate crime. It delves into how AI technologies, through advanced data analysis and predictive analytics, are revolutionizing the detection and prevention of financial fraud, insider trading, and non-compliance with environmental regulations. The integration of AI into the financial sector is emphasized as pivotal for developing robust fraud prevention algorithms, which analyze vast data volumes to identify malicious activities and prevent financial losses. Moreover, the article probes into the ethical and privacy concerns surrounding the deployment of AI, highlighting the balance needed between efficient crime prevention and safeguarding individual privacy rights. Garcia-Segura's analysis also underscores the need for international cooperation and legal adaptation to uphold ethical standards in AI's application, ensuring it promotes fairness and justice in combating corporate crime.

1. Introduction

Corporate crime, in its essence, encapsulates illegal actions undertaken by corporations or their representatives, primarily aimed at enhancing profit margins or consolidating market dominance. A critical avenue of such misconduct includes financial fraud, encompassing activities like embezzlement, insider trading, and accounting fraud (Friedrichs, 2020). Another significant facet entails environmental violations, which involve the circumvention of regulations designed to protect the environment, thereby inflicting irreversible damage upon ecosystems (Bisschop, 2017).

The digital acceleration of society led to a significant increase in cyber-enabled financial and economic crimes (Europol, 2023). Technological developments are leading to the emergence of new risks including through the growth of cybercrimes, the increasing misuse of cryptocurrencies and a new breed of sophisticated professional enablers able to create opaque structures and move money increasingly in real-time (OECD, 2021).

The implications of the surge in corporate crime are profound and multifarious. Economically, these crimes can lead to significant financial losses for stakeholders and erode public trust in markets and institutions (United Nations Office on Drugs and Crime, 2010). To this extent, according to a recent study by the Association of Certified Fraud Examiners, corporations lose approximately 5% of their annual revenue to fraud and other forms of corporate crime (Association of Certified Fraud Examiners, 2020).

Environmentally, crimes such as illegal dumping and excessive emissions have a significant impact on the health of our planet. With the increasing population and rapid industrialization, these activities have become a major threat to the delicate balance of our ecosystem. They not only devastate natural landscapes, but also damage crucial habitats for wildlife, disrupting the food chain and leading to extinction of species (Bisschop, 2017). Therefore, economic crime, environmental crime, and cybercrime are closely linked through their interconnected impact on corporate misconduct. Economic crimes, such as insider trading, accounting fraud, and embezzlement, directly manipulate financial statements and market perceptions, exemplified by the Enron scandal where fraudulent accounting practices led to the company's collapse (Rezaee, 2005). Corporations, driven by profit maximization, may also engage in environmental crimes by circumventing regulations to cut costs, such as improper chemical disposal or emission exceedances. A notable case is Volkswagen's 2015 diesel emissions scandal, where software was used to cheat on emissions tests, causing both financial and environmental harm. These corporate crimes not only facilitate an unfair market advantage but also generate substantial negative externalities, undermining regulatory frameworks intended to safeguard stakeholder interests and environmental sustainability (Eger and Schäfer, 2018).

In parallel, the rise of cybercrime has further complicated corporate crime, creating new avenues for economic and environmental malfeasance (Sutherland et al., 2019). Cyber-enabled economic crimes, like

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ransomware attacks and data breaches, can destabilize corporate operations and financial stability. For instance, the 2017 Equifax breach exposed the personal information of millions, resulting in significant financial losses and reputational damage.

Corporations are often held criminally responsible not only for committing economic crimes but also for failing to prevent them. In jurisdictions like the UK, legislation holds companies accountable for offences such as failing to prevent fraud, bribery, and tax evasion. Laws like the Bribery Act 2010 and the Criminal Finances Act 2017 underscore the corporate duty to implement adequate controls to prevent such crimes. For example, a corporation could face severe penalties if found guilty of not having sufficient anti-bribery measures in place, even if senior management was unaware of individual employees' illicit activities. Furthermore, corporations can be criminally charged for failing to implement measures to prevent money laundering or for facilitating transactions that breach financial sanctions, as highlighted in cases of international banks fined for non-compliance with anti-money laundering regulations and sanctions policies.

This accountability framework demonstrates the critical role that AI can play in preventing both internal and external misconduct, which could otherwise hold the organization responsible. To address these challenges, stringent compliance and robust monitoring mechanisms are essential.

To this extent, we have seen in the last two years significant papers published on this topic, like the "State of the art in financial statement fraud detection: A systematic review" by Shahana et al. (2023), "AI-Driven Fraud Detection Systems: A comparative Study across Banking, Insurance and Healthcare" by Pankaj Zanke (2023) and "An Analysis on Financial Statement Fraud Detection for Chinese Listed Companies Using Deep Learning" by Xiuguo and Shengyong (2022). One of the main recommendations Mr. Zanke (2023) made to stakeholders and policymakers was the following:

Foster Ethical and Responsible AI Use: Promoting ethical and responsible AI use is essential to ensure the fair and equitable treatment of individuals in fraud detection processes. Policymakers should establish guidelines, standards, and regulatory frameworks that prioritize transparency, accountability, and fairness in AI-driven fraud detection, while protecting privacy and civil liberties (Zanke, 2023, p. 19).

Finally, this article explores the novel and transformative role AI plays in enhancing corporate compliance and monitoring to mitigate corporate crime. It delves into how AI technologies, through advanced data analysis and predictive analytics, are revolutionizing the detection and prevention of financial fraud, insider trading, and non-compliance with environmental regulations. The integration of AI into the financial sector is emphasized as pivotal for developing robust fraud prevention algorithms, which analyze vast data volumes to identify malicious activities and prevent financial losses. Moreover, the article probes into the ethical and privacy concerns surrounding the deployment of AI, highlighting the balance needed between efficient crime prevention and safeguarding individual privacy rights.

A qualitative research methodology was employed to comprehensively explore the role of AI in enhancing corporate compliance and mitigating corporate crime. The study involved a systematic review of recent literature focusing on AI-driven techniques in fraud detection, compliance monitoring, and their application across various sectors, including finance and environmental regulation. Key sources included academic papers and industry reports, allowing for an extensive analysis of current AI technologies and practices. This mixed-method approach enriched the understanding of AI's transformative impact on corporate crime prevention and its associated ethical implications.

In our first section, we will see the impact the AI is having in compliance and monitoring.

2. Artificial intelligence: a game changer in compliance and monitoring

The integration of AI¹ into various sectors has significantly transformed operations, strategies, and outcomes, particularly within the realms of compliance and monitoring. AI technologies, through their ability to process and analyze vast volumes of data at unprecedented speeds, are revolutionizing the way organizations approach and enforce compliance requirements.

AI-based systems, leveraging machine learning, natural language processing, and predictive analytics, enable a more dynamic and efficacious approach to compliance monitoring. Traditional compliance methods, often manual and time-intensive, are increasingly being complemented and, in some instances, replaced by AI-driven processes. One of the fundamental advantages of AI in compliance is its predictive capability, which allows for the identification of potential risks and non-compliance issues before they materialize (Arner et al., 2020). Through continual learning and adaptation, AI systems can anticipate patterns that may indicate non-compliance, thereby enabling pre-emptive action.²

Furthermore, AI significantly enhances the efficiency of monitoring activities. By automating routine tasks, AI systems can sift through colossal datasets, identifying anomalies and flagging potential compliance issues without human fatigue or error (Bughin et al., 2017). This automation extends beyond simple rule-based tasks to more complex analyses, such as evaluating communications for potential insider trading or other unethical behaviors concealed within large volumes of data (Rai, 2020).

To this extent, the U.S. Department of Treasury (2024) asserted recently that in order to stay "ahead of technologically adept fraudsters requires proactive, AI-enhanced fraud detection methods, particularly leveraging GenAI for early identification and mitigation of fraud (p. 46)".

As we are starting to see, AI systems and tools have the potential to revolutionize the way corporations approach the prevention of corporate crime.

AI's impact on compliance is not merely operational but also strategic. It facilitates a more informed decision-making process, where compliance strategies are continuously refined based on data-driven insights. Moreover, AI's capacity for real-time monitoring ensures that compliance measures are immediately adjusted to correspond with evolving regulatory environments and organizational needs. This agility is pivotal in a landscape where regulatory requirements are continually changing, and the cost of non-compliance can be substantial.

However, the application of AI in compliance also poses new challenges, particularly concerning data privacy, ethical considerations, and the potential for systemic biases within AI algorithms (Martin, 2019). Ensuring that AI systems operate within ethical guidelines and regulatory frameworks is paramount to realizing their full potential in compliance and monitoring.

Therefore, the transformative power of AI in compliance and monitoring is undeniable, offering unprecedented efficiencies and strategic advantages. This next section will delve into how advanced data

¹ One recent definition of AI which we consider very relatable to this article is the one provided by Hussein Abbass in 2021: Artificial Intelligence is social and cognitive phenomena that enable a machine to socially integrate with a society to perform competitive tasks requiring cognitive processes and communicate with other entities in society by exchanging messages with high information content and shorter representations.

² Generative AI, sometimes called gen AI, is artificial intelligence that can create original content—such as text, images, video, audio or software code—in response to a user's prompt or request. This technology relies on sophisticated machine learning models called deep learning models—algorithms that simulate the learning and decision-making processes of the human brain (Stryker and Scapicchio, 2024).

analytics techniques facilitate early detection of compliance breaches, setting the stage for proactive risk management and mitigation.

3. Early detection through data analysis

AI, particularly machine learning (ML) and natural language processing (NLP), enables the analysis of vast amounts of corporate data at speeds and accuracies unattainable by human evaluators. By learning from historical data, AI algorithms can identify anomalies, trends, and patterns associated with fraudulent activities (Brennan, 2019). For instance, ML models trained on financial records can detect inconsistencies or irregularities that may signal embezzlement or accounting fraud. Similarly, NLP can scrutinize emails and communication logs for evidence of insider trading or ethical breaches, analyzing semantic content for suspicious behaviors (Lu et al., 2018).

Predictive analytics in AI utilizes machine learning algorithms to analyze historical data and forecast future occurrences of corporate crime. By examining previous instances of fraud, embezzlement, and other criminal activities, AI systems can learn to identify the precursors to such actions (Chui et al., 2018). For example, AI can scrutinize financial transactions for patterns indicative of embezzlement or assess communication data for signs of insider trading, thereby enabling organizations to intervene before the crime is perpetrated. The efficacy of predictive analytics stems from its ability to continuously refine its models based on new data, thus improving its predictive accuracy over time (O'Neil, 2016).

According to the 2024 Anti-Fraud Technology Benchmarking Report published by the Association of Certified Fraud Examiners (2024), more than 90 % of the organizations that participated in the study use some sort of data analysis as part of their anti-fraud program. AI and ML have the greatest adoption rate, with nearly one-third of organizations that do not currently use the technology expecting to add it to their anti-fraud program in the near future. This means that by 2026, half of all organizations expect to use AI and ML as part of their fraud analytics initiatives.

Furthermore, the 2023 Fraud and Financial Crime Report published by Kroll (2023), had the following conclusions:

- Investments in technology, increase in cybersecurity budgets and undertaking more frequent business risk assessments have been cited by respondents as the three main steps that firms intend to take during the next year to combat the increase in financial crime.
- To counter a potential uptick in financial crime risks, two-thirds of respondents globally are planning to invest more in technology, with nearly half of respondents citing data integrity as the biggest challenge when implementing new technologies.

The report highlighted the need for increased collaboration between financial institutions and law enforcement agencies. This is especially important considering that a significant portion of financial crimes are attributed to organized crime networks (Leukfeldt et al., 2016). Under these circumstances, the integration of AI into compliance and monitoring systems provides a proactive approach to corporate governance. Traditionally, corporate crime detection has relied on retrospective analysis and whistleblower reports (Ivancevich et al., 2003). However, AI's predictive analytics capability allows for the early identification of potential risks, facilitating timely interventions (O'Neil, 2016).

Furthermore, the dynamic nature of corporate crime, with perpetrators continually evolving their strategies to circumvent detection, means that AI systems must be adaptable and continuously updated to reflect changing tactics and regulations (Scherer, 2016). This requires a commitment to ongoing learning and development of AI models, coupled with expert oversight to interpret AI-generated insights and implement appropriate responses.

As mentioned, the agility and adaptability of AI systems are vital in maintaining their effectiveness against the evolving strategies of

corporate criminals. Moving forward, the next section will highlight how advancements in AI and machine learning are specifically tailored to detect and prevent fraudulent activities within the financial sector, ensuring greater security and integrity.

4. Financial sector innovations: fraud prevention algorithms

The financial sector stands at the crossroads of rapid technological advancements and the enduring challenge of fraud prevention. As financial transactions grow in volume and complexity, traditional manual methods of fraud detection have become inadequate, necessitating the development and implementation of innovative technological solutions (Kou et al., 2004).

The proliferation of big data analytics has further amplified the effectiveness of fraud prevention algorithms. With the ability to process and analyze vast datasets, financial institutions can uncover complex fraud schemes that may span across multiple accounts and transactions. This big data approach allows for the identification of subtle correlations and patterns that human analysts might overlook, offering a macroscopic view of fraudulent activities across the financial ecosystem.

Fraud prevention algorithms operate by analyzing patterns and anomalies in transaction data, flagging activities that deviate from established behavioral models as potentially fraudulent. These algorithms leverage a vast array of data sources, including transaction histories, customer profiles, and external databases, to generate comprehensive risk assessments. The use of ML models, in particular, allows these systems to continuously refine their detection capabilities based on new data, thereby enhancing their predictive accuracy over time (Bhattacharyya, 2011).

One of the foremost applications of fraud prevention algorithms is in credit card fraud detection. Algorithms analyze transaction data in real-time, assessing variables such as transaction amount, location, and frequency against known fraud indicators and historical account activity. Suspicious activities trigger alerts that can temporarily freeze transactions, pending further investigation. This real-time analysis and response capability significantly reduce the financial losses associated with credit card fraud (Dal Pozzolo et al., 2017).

As of this moment, both Mastercard and Visa have recently unveiled AI and ML fraud detection internal applications to consumers. The Mastercard product is called "Decision Intelligence" and according to Ajay Bhalla, president of Cyber and Intelligence at the company, this tool has the ability to assess relationships between multiple entities surrounding a transaction to determine its risk, in less than 50 ms, boosting fraud detection rates on average by 20 % (Bhalla, 2024).

Visa, on their part, reported that their tool "Fraud Management Essentials", an integrated ML model, generates a highly accurate risk score for every transaction in less than a second, helping detect and prevent fraud faster than a human ever could (Barding, 2023).

Similarly, fraud prevention algorithms play a critical role in detecting and preventing unauthorized access to accounts and phishing attempts. By analyzing login patterns, device information, and transaction behaviors, these systems can identify irregularities that may indicate an account compromise, providing an additional layer of security in the digital financial landscape.

The arms race between fraudsters and financial institutions continues, as perpetrators constantly develop new tactics to circumvent detection. Additionally, the risk of false positives, where legitimate transactions are mistakenly flagged as fraudulent, can disrupt customer experiences and operational efficiency. Ongoing research and development efforts aimed to address these challenges by enhancing the precision of fraud detection algorithms and minimizing erroneous alerts needs to be carried out.

In such manner, let's look at what British authorities have come up with recently to improve fraud detection. The UK's Payment Systems Regulator (PSR) announced it would be introducing a new, mandatory

requirement for all UK Payment Service Providers to reimburse their customers who become victims of APP fraud, beginning on October 2024. However, it is important to note that a particular level of consumer caution will be applied, which may result in the denial of reimbursement claims. This decision will be based on the provider's ability to demonstrate that the customer did not exercise sufficient care or cooperation. It is therefore the customer's responsibility to pay attention to any warnings issued by their bank regarding suspected APP fraud attempts, to promptly inform their bank of any fraudulent activity, to share relevant information with their bank, and to consent to the reporting of the fraud to the police (Lovell, 2024).

Building on this idea, the next section delves into ethical and privacy concerns relating to AI systems for preventing crime in general.

5. Ethical considerations and privacy concerns

The deployment of AI in corporate crime prevention is not without challenges. Data privacy concerns emerge as AI systems require access to sensitive and proprietary information. Ensuring the protection of such data while allowing for effective analysis is a critical consideration. Moreover, the potential for AI bias, where algorithms might inadvertently discriminate based on flawed input data, necessitates rigorous testing and validation to ensure fairness and accuracy in detection efforts (Raj, 2020).

The risk of bias in AI algorithms—stemming from skewed or incomplete training data—can lead to inaccurate risk assessments or unjustly targeted individuals (Martin, 2019). Moreover, the dynamic nature of corporate crime requires that AI models are continuously updated to adapt to new methods of criminal behavior, necessitating ongoing investment in AI systems and data analytics expertise.

Therefore, a productive use of AI in predictive crimes must always be dependent on a human rights compliant use of AI which keeps in mind the following critical areas: transparency, accountability and bias (European Crime Prevention Network, 2022).

Precisely transparency and accountability are some of the principles included in the OECD Principles on AI: transparency should provide meaningful information that enables those adversely affected by an AI system to challenge its output (for example, employees of a company that has an internal AI fraud prevention system) and accountability ensures that AI enabling companies and organizations should apply a systemic risk management approach to each phase of the AI system lifecycle on an ongoing basis (OECD, 2024).

One of the primary ethical considerations is the privacy of individuals. The processing and analysis of personal data raises concerns over privacy breaches and the unauthorized use of personal information (Martin, 2019). For instance, surveillance technologies powered by AI, used to monitor employee communications or financial transactions, may inadvertently intrude on personal privacy, leading to ethical dilemmas over the balance between effective crime prevention and the right to privacy (Mittelstadt, 2016). If we add this to the equation the factor of bias, it can ultimately lead to disproportionate scrutiny or wrongful accusations against individuals based on flawed algorithmic assessments (Barocas and Selbst, 2016).

Consequently, the adaptability of legal frameworks is crucial and there is a need for international cooperation and consensus-building regarding the use of AI in crime prevention. Criminal activities, especially those facilitated by technology, often transcend national borders, necessitating a collaborative approach to developing legal and regulatory standards that are globally applicable and enforceable (Taddeo and Floridi, 2018). Any global framework must ensure that AI-driven crime prevention strategies uphold human rights and do not exacerbate existing inequalities or biases within criminal justice systems (Cath, 2018).

While there are various legal instruments which could be applied to AI in law enforcement, it is essential to have a comprehensive legal framework for the development and use of AI systems in general, and for law enforcement specifically (Kaferanis et al., 2023).

In a Policy Paper published in 2022, the International NGO "Fair Trials" had this to say about AI in criminal justice systems:

AI systems have been shown to directly generate and reinforce discriminatory and unjust outcomes; infringing fundamental rights, they have been found to have little to no positive influence on the quality of human decisions, and they have been criticized for poor design that does not comply with human rights standards (Fair Trials, 2022, p. 1).

This leads us to a reevaluation and potential reshaping of legal standards to ensure that the use of AI in crime prevention aligns with fundamental legal principles and societal values. The future direction of AI-driven crime prevention strategies will likely hinge on achieving a harmonious balance between technological capabilities and legal and ethical norms, which are in still in development, as we speak.

For example, the European Union has been at the forefront of establishing comprehensive frameworks to regulate AI technologies, with the proposed EU AI Act setting a precedent. This regulation aims to provide a harmonized approach to AI deployment across member states, ensuring that AI systems used, including those for crime prevention, are aligned with EU values and fundamental rights such as privacy, non-discrimination, and data protection. The act categorizes AI applications into risk levels, ranging from minimal to high, with stringent obligations on systems identified as high-risk, including those in law enforcement. As the EU aims to foster trust in AI, it emphasizes promoting innovation while safeguarding societal interests, potentially setting global standards. This regulatory approach might exert significant influence on AI strategies worldwide, pushing for ethical AI deployment and possibly encouraging other nations to adopt similar measures.

Collaboration among international bodies, such as the United Nations, INTERPOL, and various regional organizations, is paramount to developing and implementing global standards for AI-driven crime prevention. These entities can play a pivotal role in facilitating dialogue, sharing best practices, and coordinating efforts to create a unified approach to leveraging AI technologies. Establishing international agreements and frameworks that support cooperation and mutual assistance in AI-driven crime prevention efforts is critical for their success across borders.

Navigating the ethical dimensions of AI deployment in crime prevention presents a complex array of challenges that necessitate careful consideration and action. The tension between privacy concerns and the need for effective crime detection underscores the critical importance of developing AI frameworks that are both transparent and accountable. This discussion leads us to the overall conclusions of our exploration, where we synthesize these insights to outline potential pathways for the responsible advancement of AI technologies in crime prevention.

6. Conclusion

The deployment of AI in preventing corporate crime presents a transformative opportunity for businesses to enhance their compliance and monitoring capabilities, fundamentally altering the landscape of corporate governance. As stated in the article, AI's predictive analytics, data processing, and algorithmic assessments provide an advanced forefront against financial fraud, embezzlement, cybercrimes, and other forms of corporate malfeasance. These technologies enable early detection through extensive data analysis, spotting anomalies that may indicate fraudulent activities far more efficiently than traditional methods. Moreover, the advent of fraud prevention algorithms specifically within the financial sector showcases AI's significant contribution to safeguarding transactions and protecting stakeholders' interests.

However, the incorporation of AI into corporate crime prevention strategies is not without its ethical and privacy concerns. The balance between effective crime prevention and the preservation of individual

privacy rights emerges as a paramount consideration. AI systems, especially those involved in surveillance and monitoring activities, pose a risk of infringing on privacy and inappropriately utilizing personal information. Additionally, the potential for algorithmic bias, as noted in the documents, highlights the critical need for transparency, accountability, and fairness in AI applications. These concerns necessitate stringent ethical guidelines and comprehensive legal frameworks to ensure AI technologies are used responsibly and equitably.

The future direction of AI-driven corporate crime prevention strategies will undeniably hinge on achieving a harmonious balance between technological capabilities and adherence to legal and ethical norms. International cooperation and consensus-building regarding the use of AI in crime prevention are crucial, given the transnational nature of many corporate crimes and the global operation of corporations. Leveraging AI's full potential will require a collaborative effort among governments, international bodies, and the private sector to develop and implement globally applicable and enforceable standards that safeguard human rights and promote fairness and equality within the justice system.

Furthermore, the ongoing development and refinement of AI technologies must prioritize enhancing the precision of fraud detection while minimizing false positives, which could disrupt legitimate business activities and customer relations. Investing in AI and machine learning for fraud analytics initiatives, as noted in the documents, signifies a proactive approach to mitigating financial crime risks. This investment, coupled with increased collaboration between financial institutions and law enforcement, indicates a committed stride toward innovating more resilient and responsive systems for corporate governance.

In conclusion, AI's role in preventing corporate crime offers promising avenues for significantly improving compliance and monitoring efforts, marked by enhanced efficiency, predictive accuracy, and strategic insights. Nevertheless, the challenges of privacy concerns, potential biases, and the need for legal adaptability call for a nuanced approach to AI implementation. By fostering ethical and responsible AI use, establishing robust legal frameworks, and promoting international collaboration, the shift toward AI-enhanced corporate crime prevention can lead to more secure, trustworthy, and equitable corporate environments.

CRedit authorship contribution statement

Luis A. Garcia-Segura: Writing – review & editing, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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